

PredictFun: A No-Subsidy Prediction Market Launchpad

PredictFun
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Abstract

This paper specifies a prediction market launchpad in which binary markets form through a pre-graduation LMSR pricing curve without protocol subsidy or guaranteed fill. Markets can form real betting activity immediately, but pre-graduation bets represent intent rather than guaranteed fill; only cleared exposure becomes a final Conditional Token Framework (CTF)-style outcome-token claim. A market graduates only when enough compatible opposing demand exists to form a fully collateralized settlement pool. At graduation, excess exposure on the overbet side is pared down until every retained bet is matched by opposing collateral; removed exposure is refunded, and the matched remainder converts into fully backed outcome tokens.

The design combines the continuous price discovery of LMSR with fixed-payout batch clearing. LMSR gives every market a smooth probability curve before graduation. Batch clearing prevents insolvency by ensuring that final redeemable claims are backed by real collateral rather than virtual liquidity. Resolution is handled separately through an optimistic process: an AI resolver proposes an outcome with evidence, bonded disputes can challenge it, and high-value or contested cases escalate to human or UMA-style arbitration.

Graduation is the transition from intent to CTF-style ownership. Before graduation, participants hold provisional receipts priced by LMSR; after graduation, matched exposure becomes fully backed CTF outcome tokens. Those tokens have fixed redemption semantics: complete outcome sets are collateral-backed, winning outcomes redeem at resolution, and secondary trading happens over backed claims rather than provisional receipts.

1. Introduction

Prediction market launchpads inherit a tension from two different market types.

Token launchpads use bonding curves to create immediate trading. The curve can quote a token price before deep two-sided liquidity exists because the token has no fixed redemption value.

Prediction markets are different. A YES share or NO share is a contingent claim:

YES $\mapsto$ 1	if the event resolves YES
NO $\mapsto$ 1	if the event resolves NO

For a binary fixed-payout market, a collateral-complete pair satisfies:

$$1 \text{ YES} + 1 \text{ NO} = 1 \text{ unit of collateral}$$

If a mechanism sells 100 YES at a price of 0.30, it collects 30 units of collateral. If YES wins, those shares owe 100 units. Without a subsidy, LP inventory, or compatible NO demand, the missing 70 units are an insolvency.

This paper proposes a no-subsidy alternative:

Before graduation: LMSR-priced provisional receipts.

At graduation: fixed-payout batch clearing and refunds.

After graduation: retained CTF-style outcome tokens.

The mechanism does not promise instant fill. It promises instant participation in price discovery, then deterministic clearing once the market has enough real opposing demand.

2. Design Goals

The mechanism is designed around five goals:

1. **No protocol subsidy.** The launchpad should not need to bankroll the missing side of early prediction bets.
2. **Continuous pre-grad pricing.** Bettors should see a live probability curve rather than a static pool.
3. **No hidden insolvency.** A final fixed-payout claim should never exist unless it can be fully backed.
4. **Locked or improved fill odds.** If a provisional receipt survives graduation, its execution price should be no worse than the bettor's entry price.
5. **Resolution independence.** Trading mechanics and truth resolution should be separate layers.

The design intentionally omits trading fees. Fees can be added later, but introducing them before the fill/refund mechanism is settled makes user accounting harder to reason about.

3. Background: LMSR, CTF Outcome Tokens, And Pari-Mutuel Markets

LMSR

The Logarithmic Market Scoring Rule, or LMSR, gives the pre-graduation market a smooth probability curve whose marginal prices move continuously with YES and NO demand.

For a binary market:

$$C(q_{\text{yes}}, q_{\text{no}}) = b \ln \left(e^{q_{\text{yes}}/b} + e^{q_{\text{no}}/b} \right)$$

The marginal YES price is:

$$P_{\text{yes}} = \frac{e^{q_{\text{yes}}/b}}{e^{q_{\text{yes}}/b} + e^{q_{\text{no}}/b}}$$

This follows by differentiating the cost function:

$$\begin{aligned} \frac{dC}{dq_{\text{yes}}} &= b \left(\frac{1}{e^{q_{\text{yes}}/b} + e^{q_{\text{no}}/b}} \right) \left(\frac{1}{b} \right) e^{q_{\text{yes}}/b} \\ &= \frac{e^{q_{\text{yes}}/b}}{e^{q_{\text{yes}}/b} + e^{q_{\text{no}}/b}} \end{aligned}$$

The marginal NO price is:

$$P_{\text{no}} = 1 - P_{\text{yes}}$$

The cost of buying additional provisional shares is:

$$\text{cost} = C(q_{\text{after}}) - C(q_{\text{before}})$$

Equivalently, the trade cost is the integral of the marginal LMSR price over the trade path:

$$\begin{aligned} \text{cost}_{\text{yes}}(s) &= \int_0^s P_{\text{yes}}(q_{\text{yes}} + x, q_{\text{no}}) dx \\ &= C(q_{\text{yes}} + s, q_{\text{no}}) - C(q_{\text{yes}}, q_{\text{no}}) \end{aligned}$$

In a conventional LMSR automated market maker, the liquidity parameter b implies a bounded market-maker loss. That model requires a funded subsidy. In this design, b has a different role: it controls the smoothness of pre-graduation pricing and how strongly early demand moves the displayed probability.

CTF Outcome Tokens

In a CTF-style prediction market, the payoff of a retained position is known:

$$\begin{array}{ll} 1 \text{ retained YES} \mapsto 1 & \text{if YES resolves true} \\ 1 \text{ retained NO} \mapsto 1 & \text{if NO resolves true} \end{array}$$

This is the market structure users expect from Polymarket-style YES/NO trading. The user does not merely own a pro-rata claim on a pot. They own a side-specific outcome token with a fixed terminal payout.

For a no-subsidy launchpad, this creates a constraint:

retained YES exposure and retained NO exposure must be jointly collateral-complete.

Pari-Mutuel Contrast

In a pure pari-mutuel market, winners share the pool and final odds are not fully known until betting closes. That is solvent, but it is not the same as a fixed-payout YES/NO market.

PredictFun uses a related solvency discipline but not the same terminal payoff rule. Excess demand is refunded before settlement. The remaining demand converts into CTF-style fixed-payout outcome tokens.

The hybrid can be summarized as:

LMSR for pre-grad price discovery.
Batch clearing for CTF-style outcome-token fill.
Optimistic oracle for truth.

4. Mechanism Overview

Each market has two phases.

Pre-Graduation

Bettors buy provisional YES or NO receipts against an LMSR curve. Each receipt records:

side
provisional shares
cost basis
average entry price

A receipt is not a final redeemable position. It is a claim to participate in graduation clearing.

The displayed LMSR quote remains useful:

price $\approx$ implied probability

But the quote is not a fill guarantee. It is the maximum execution price for any retained portion of the receipt.

Graduation

The market graduates when compatible YES/NO demand reaches a threshold. Compatibility means the two sides can jointly fund CTF-style outcome-token claims:

$$p_{\text{yes}} + p_{\text{no}} \geq 1$$

At graduation:

1. New pre-grad betting stops.
2. YES and NO receipt lots are aggregated by price.
3. Compatible YES and NO lots are matched.
4. Unmatched and overbet exposure is refunded.

- Retained exposure becomes CTF-style YES/NO outcome tokens.
- The market continues in a mature venue or proceeds to resolution.

5. Pre-Graduation Receipts

A pre-graduation receipt is the accounting object created when a bettor moves the LMSR state. It is not a CTF token yet and it does not by itself represent a redeemable claim. It records the bettor's priced intent so graduation clearing can later decide how much of that intent is backed by compatible opposing demand.

The receipt variables come from three places:

VARIABLE	SOURCE	MEANING
b	Market parameter	Controls LMSR curve smoothness and price sensitivity.
$q_{\text{yes}}, q_{\text{no}}$	LMSR state	Running score state used to quote marginal prices.
s_l	Trade size	Provisional shares bought in receipt lot l .
c_l	LMSR cost difference	Collateral paid for receipt lot l .
p_l	Derived from receipt	Average entry price, equal to c_l / s_l .
Y, N	Receipt aggregates	Total provisional YES and NO shares.

For each receipt lot l , the market tracks:

$$\begin{aligned}
 \text{side}_l &\in \{\text{YES}, \text{NO}\} \\
 s_l &= \text{provisional shares} \\
 c_l &= \text{cost basis} \\
 p_l &= \frac{c_l}{s_l}
 \end{aligned}$$

The aggregate quantities are:

$$\begin{aligned}
 Y &= \sum \text{YES shares} \\
 N &= \sum \text{NO shares} \\
 C_y &= \sum \text{YES cost basis} \\
 C_n &= \sum \text{NO cost basis}
 \end{aligned}$$

The displayed LMSR state can use Y and N directly:

$$\begin{aligned}
 q_{\text{yes}} &= Y \\
 q_{\text{no}} &= N
 \end{aligned}$$

More generally, a market can include an initial pricing offset:

$$q_{\text{yes}} = q_{\text{yes},0} + Y$$

$$q_{\text{no}} = q_{\text{no},0} + N$$

For a neutral market, $q_{\text{yes},0} = q_{\text{no},0} = 0$, so the initial price is 0.50. For a market that should open at initial probability P_0 , only the difference between the offsets matters:

$$P_0 = \frac{e^{q_{\text{yes},0}/b}}{e^{q_{\text{yes},0}/b} + e^{q_{\text{no},0}/b}}$$

$$q_{\text{yes},0} - q_{\text{no},0} = b \ln \left(\frac{P_0}{1 - P_0} \right)$$

The offset is still just LMSR state. It is not a subsidy and does not create backed exposure.

Given a current state $(q_{\text{yes}}, q_{\text{no}})$, a YES receipt for s_l provisional shares is priced as:

$$q_{\text{before}} = (q_{\text{yes}}, q_{\text{no}})$$

$$q_{\text{after}} = (q_{\text{yes}} + s_l, q_{\text{no}})$$

$$c_l = C(q_{\text{after}}) - C(q_{\text{before}})$$

$$p_l = \frac{c_l}{s_l}$$

A NO receipt is symmetric:

$$q_{\text{before}} = (q_{\text{yes}}, q_{\text{no}})$$

$$q_{\text{after}} = (q_{\text{yes}}, q_{\text{no}} + s_l)$$

$$c_l = C(q_{\text{after}}) - C(q_{\text{before}})$$

$$p_l = \frac{c_l}{s_l}$$

So p_l is the average LMSR price paid across the trade, not necessarily the final marginal quote after the trade. This matters because graduation clearing uses the collateral actually paid per provisional share.

If the user enters by budget instead of share amount, the share amount can be solved from the same LMSR cost equation. For a YES budget c , define:

$$A = e^{q_{\text{yes}}/b}$$

$$B = e^{q_{\text{no}}/b}$$

Then:

$$c = b \ln \left(\frac{Ae^{s/b} + B}{A + B} \right)$$

$$e^{c/b}(A + B) = Ae^{s/b} + B$$

$$e^{s/b} = \frac{e^{c/b}(A + B) - B}{A}$$

$$s = b \ln \left(\frac{e^{c/b}(A + B) - B}{A} \right)$$

For a NO budget, swap **A** and **B**:

$$s = b \ln \left(\frac{e^{c/b}(A + B) - A}{B} \right)$$

Receipts can be bucketed by side and entry price. That gives the clearing algorithm a way to preserve price discipline without tracking an order book forever.

6. Compatibility And Filled Market Cap

A YES lot with entry price **p_y** and a NO lot with entry price **p_n** can be converted into CTF-style outcome-token claims only if:

$$p_y + p_n \geq 1$$

This condition is not an LMSR condition. LMSR determines **p_y** and **p_n** by pricing each receipt when it is created. The compatibility condition comes from the CTF collateral requirement:

$$1 \text{ YES outcome token} + 1 \text{ NO outcome token} = 1 \text{ unit of collateral}$$

If the clearing engine keeps **f** YES shares from one receipt and **f** NO shares from another receipt, the two receipts contribute:

$$\begin{aligned} \text{YES collateral} &= fp_y \\ \text{NO collateral} &= fp_n \\ \text{total contributed} &= f(p_y + p_n) \end{aligned}$$

The CTF-style complete set requires:

$$\text{required collateral} = f$$

Therefore:

$$\begin{aligned} f(p_y + p_n) &\geq f \\ p_y + p_n &\geq 1 \end{aligned}$$

If:

$$p_y + p_n < 1$$

then the two lots do not contain enough collateral to mint one YES and one NO per retained share without subsidy. They cannot be matched as fixed-payout exposure.

For a compatible pair, the fill amount is:

$$f = \min(\text{YES shares remaining}, \text{NO shares remaining})$$

The retained collateral contributed by the pair is:

$$\text{collateral}_{\text{pair}} = f(p_y + p_n)$$

The collateral needed to back the fixed-payout pair is:

$$\text{required}_{\text{pair}} = f$$

Any surplus is:

$$\text{surplus}_{\text{pair}} = f(p_y + p_n - 1)$$

With no fees, the surplus should be returned as price improvement or explicitly routed into post-graduation liquidity. The simplest paper model returns it.

The filled market cap is the total compatible matched amount:

$$F = \sum \text{matched } f$$

A market graduates when:

$$F \geq \text{graduation}_{\text{threshold}}$$

7. Graduation Clearing

Graduation clearing is a batch auction over receipt lots.

A simple v1 clearing rule:

1. Sort YES lots from highest p_y to lowest.
2. Sort NO lots from highest p_n to lowest.
3. Match lots while $p_y + p_n \geq 1$.
4. For each match, retain $f = \min(\text{yes_remaining}, \text{no_remaining})$.
5. Refund unmatched shares at their cost basis.
6. Return any surplus as price improvement.

This rule favors the most price-compatible demand. It is not first-in-first-out. It is closer to a batch clearing market: the receipts most willing to fund fixed-payout exposure are retained first.

For a retained YES amount f from a lot with entry price p_y :

$$\begin{aligned}\text{retained}_{\text{yes}} &= f \\ \text{max retained cost} &= fp_y\end{aligned}$$

For a retained NO amount f from a lot with entry price p_n :

$$\begin{aligned}\text{retained}_{\text{no}} &= f \\ \text{max retained cost} &= fp_n\end{aligned}$$

If there is surplus:

$$\text{effective}_{\text{cost}} \leq \text{max}_{\text{retained cost}}$$

So retained bettors receive CTF-style outcome-token claims at an execution price no worse than their pre-grad entry price.

8. Refunds

For each receipt lot:

$$\begin{aligned}\text{removed}_{\text{shares}} &= \text{provisional}_{\text{shares}} - \text{retained}_{\text{shares}} \\ \text{removed}_{\text{refund}} &= \text{removed}_{\text{shares}} \cdot \text{entry}_{\text{price}}\end{aligned}$$

If the lot received price improvement from surplus:

$$\text{surplus}_{\text{refund}} = \text{retained}_{\text{shares}} \cdot \text{price}_{\text{improvement}}$$

Total refund:

$$\text{refund} = \text{removed}_{\text{refund}} + \text{surplus}_{\text{refund}}$$

The user-level invariants:

$$\begin{aligned} 0 &\leq \text{refund} \leq \text{cost}_{\text{basis}} \\ \text{retained}_{\text{cost}} &\leq \text{retained}_{\text{shares}} \\ \text{effective}_{\text{retained price}} &\leq \text{entry}_{\text{price}} \end{aligned}$$

The market-level invariant:

$$\text{total}_{\text{collateral locked}} = F$$

where F is the retained fixed-payout market cap.

9. Worked Example

Suppose three bettors enter pre-graduation:

A: 100 provisional YES at 0.30, cost basis 30

B: 100 provisional YES at 0.40, cost basis 40

C: 10 provisional NO at 0.60, cost basis 6

Compatibility:

$$\begin{aligned} A + C &= 0.30 + 0.60 = 0.90 && \text{not compatible} \\ B + C &= 0.40 + 0.60 = 1.00 && \text{compatible} \end{aligned}$$

The clearing engine matches B with C:

$$f = \min(100, 10) = 10$$

Retained exposure:

B retains 10 YES

C retains 10 NO

Refunds:

$$\begin{aligned} A_{\text{refund}} &= 100 \cdot 0.30 = 30 \\ B_{\text{removed shares}} &= 90 \\ B_{\text{refund}} &= 90 \cdot 0.40 = 36 \\ C_{\text{refund}} &= 0 \end{aligned}$$

Collateral locked:

$$\begin{aligned}B_{\text{retained cost}} &= 10 \cdot 0.40 = 4 \\C_{\text{retained cost}} &= 10 \cdot 0.60 = 6 \\ \text{total}_{\text{locked}} &= 10\end{aligned}$$

If YES resolves true:

B receives 10
C receives 0

If NO resolves true:

B receives 0
C receives 10

The 0.30 YES receipt did not fill because it lacked compatible opposing collateral. This is the correct result for a no-subsidy fixed-payout market. The bettor did not lose; the receipt was refunded.

10. Post-Graduation Trading

After graduation, retained exposure can trade as CTF-style fixed-payout outcome tokens.

A mature post-graduation venue can support:

- Trading between outcome tokens and collateral.
- Trading directly between outcome tokens.
- Splitting collateral into complete YES/NO outcome sets.
- Merging complete YES/NO outcome sets back into collateral.
- AMM or order-book liquidity over backed outcome tokens.

The important design point is that this venue starts from CTF-backed exposure. It is not responsible for rescuing undercollateralized pre-grad receipts.

11. Comparison To Alternative Designs

MECHANISM	STRENGTH	WEAKNESS	WHY NOT DEFAULT
Subsidized LMSR	Instant fills and clean probabilities	Requires protocol bankroll and can be farmed by informed flow	Violates no-subsidy goal
Pure pari-mutuel	Always solvent	Winners share a pot; final odds are not fixed at entry	Not the desired CTF-style payoff
Order book before graduation	Precise limit matching	Sparse early liquidity, complex UX	Feels slow for new markets
Constant-product AMM	Familiar AMM form	LPs take outcome risk	Needs LP capital before demand is proven
LMSR-priced fixed-payout clearing	Smooth provisional pricing and fixed-payout settlement	No guaranteed fill, refunds can surprise users	Chosen v1 tradeoff

12. Settlement And Resolution

Trading mechanics determine which claims are retained. Resolution determines which retained claims win.

Resolution should be optimistic:

1. An AI resolver proposes YES, NO, CANCEL, or another allowed result.
2. The proposal includes an evidence bundle.
3. A challenge window opens.
4. If nobody disputes, the proposal finalizes.
5. If disputed, the case escalates to human review.
6. High-value, conflicted, or appealed cases escalate to a final arbitration layer such as UMA or a similar optimistic oracle backstop.

The resolution layer should only operate on retained backed exposure:

$$\text{economic}_{\text{exposure}} = F$$

Removed and refunded exposure has no resolution rights.

13. Market Rule Schema

Every market should have a canonical rule spec:

MARKET RULE SCHEMA

```
{
  "question": "Will X happen by DATE?",
  "outcomes": ["YES", "NO"],
  "yes_criteria": "Exact condition for YES.",
  "no_criteria": "Exact condition for NO.",
  "end_time": "2026-12-31T23:59:59Z",
  "resolution_deadline": "2027-01-02T23:59:59Z",
  "primary_sources": [],
  "fallback_sources": [],
  "source_priority_rule": "Use the first source in priority order that publishes a definitive result.",
  "too_early_rule": "A proposal before the required source publishes is invalid.",
  "cancellation_rule": "Cancel only under explicit invalid-market conditions.",
  "evidence_requirements": ["url", "retrieval_timestamp", "claim_summary", "archive_hash"]
}
```

The rule spec exists to prevent the market maker from becoming the court. Trading, clearing, and resolution are separate concerns.

14. Properties

No Subsidy

The mechanism never requires a protocol-funded market-maker loss budget. Incompatible exposure is refunded instead of subsidized.

Continuous Price Discovery

LMSR gives every pre-grad market an immediately visible price path. Even before final settlement liquidity exists, the market can communicate crowd belief.

Fixed-Payout Settlement

The retained post-graduation position is a YES or NO claim with a fixed terminal payout. It is not merely a pro-rata claim on a shared pot.

Solvency By Construction

Only compatible matched exposure survives graduation. Since each retained YES/NO pair contains at least one unit of collateral, final claims can be backed.

Refundable Overbet Demand

Excess one-sided demand does not create bad debt. It creates refunds.

Resolution Modularity

The resolution layer can improve over time without changing the pre-grad clearing math.

15. Risks

Fill Misunderstanding

Users may interpret provisional receipts as guaranteed positions. The UI must label them as provisional and show estimated retained shares and refunds.

One-Sided Markets

Very popular one-sided markets can create large refund events. This is economically safe but can disappoint users.

Price-Incompatible Demand

A market can have large nominal YES and NO demand but little compatible demand if the two sides' entry prices do not sum to at least 1. Graduation should depend on compatible filled market cap, not headline provisional volume.

Clearing Rule Complexity

Batch clearing must be simple, deterministic, and heavily tested.

Manipulated Graduation

Attackers may attempt to create just enough compatible opposite-side demand to graduate a market on favorable terms. Graduation thresholds should include unique bettor counts, concentration checks, and resolution-quality gates.

Resolution Failure

If the oracle layer proposes the wrong result or resolves too early, retained positions can settle incorrectly. This is mitigated by evidence bundles, liveness windows, bonded disputes, human review, and final arbitration.

Ambiguous Markets

No clearing mechanism fixes ambiguous market wording. Market review and source-bound rules are required before pre-grad betting begins.

16. Conclusion

This paper specifies a no-subsidy prediction market launchpad that separates demand formation from final fill. LMSR provides continuous pre-graduation pricing. Fixed-payout batch clearing determines how much exposure can safely become final. Incompatible and overbet exposure is refunded, so the protocol never needs to invent missing collateral.

The result is a market that can launch quickly, show a live probability curve, and remain solvent without a market-maker subsidy. The cost is explicit: pre-graduation bets are provisional. Fixed-payout exposure is earned at graduation by the presence of compatible opposing demand.

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